Lecture 12: Cognitive Architectures

COMP3018/COMP5018: Human-Robot Interaction

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Introduction

- **Today's Topics**

  1- Cognitive architectures

- **Session Learning Outcomes**

  By the end of today's lecture you will be able to:

  1- Describe cognitive architectures and the different categories of architectures.

Example Cognitive Architectures

Surveys:

Biologically Inspired Cognitive Architectures Society, Comparative Repository of Cognitive Architectures, http://bicasociety.org/cogarch/architectures.htm (25 cognitive architectures)

A Survey of Cognitive and Agent Architectures, University of Michigan, http://ai.eecs.umich.edu/cogarch0/ (12 cognitive architectures)

W. Duch, R. J. Oentaryo, and M. Pasquier. "Cognitive Architectures: Where do we go from here?", Proc. Conf. Artificial General Intelligence, 122-136, 2008. (17 cognitive architectures)

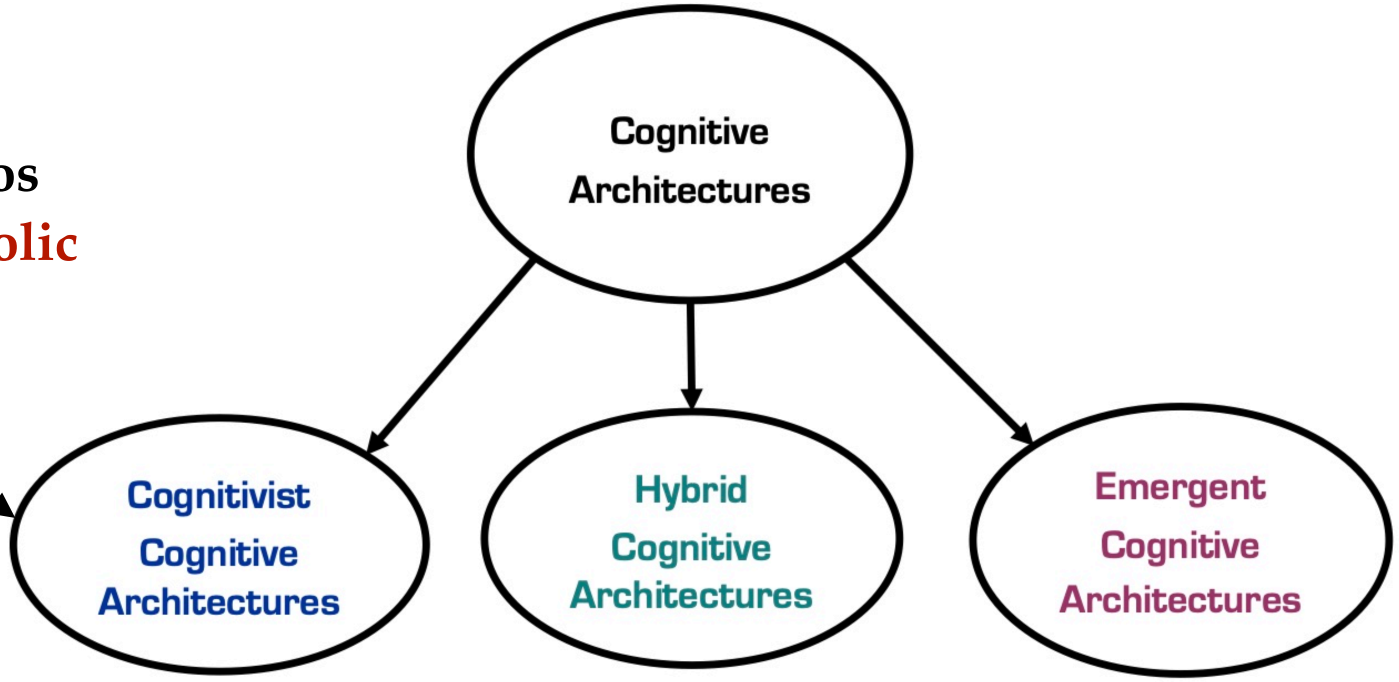
D. Vernon, G. Metta, and G. Sandini, "A Survey of Artificial Cognitive Systems: Implications for the Autonomous Development of Mental Capabilities in Computational Agents", IEEE Transactions on Evolutionary Computation, Vol. 11, No. 2, pp. 151-180, 2007. (14 cognitive architectures)

D. Vernon, C. von Hofsten, and L. Fadiga. "A Roadmap for Cognitive Development in Humanoid Robots", Cognitive Systems Monographs (COSMOS), Vol. 11, Springer, 2010. Chapter 5 and Appendix I (20 cognitive architectures)

I. Kotseruba and J. Tsotsos. 40 years of cognitive architectures: core cognitive abilities and practical applications. Artificial Intelligence Review, Vol. 53, No. 1, pp. 17-94, 2020. (84 cognitive architectures)

Cognitive Architectures

Kotseruba and Tsotsos refer to these as Symbolic Architectures



Framework in which to embed knowledge

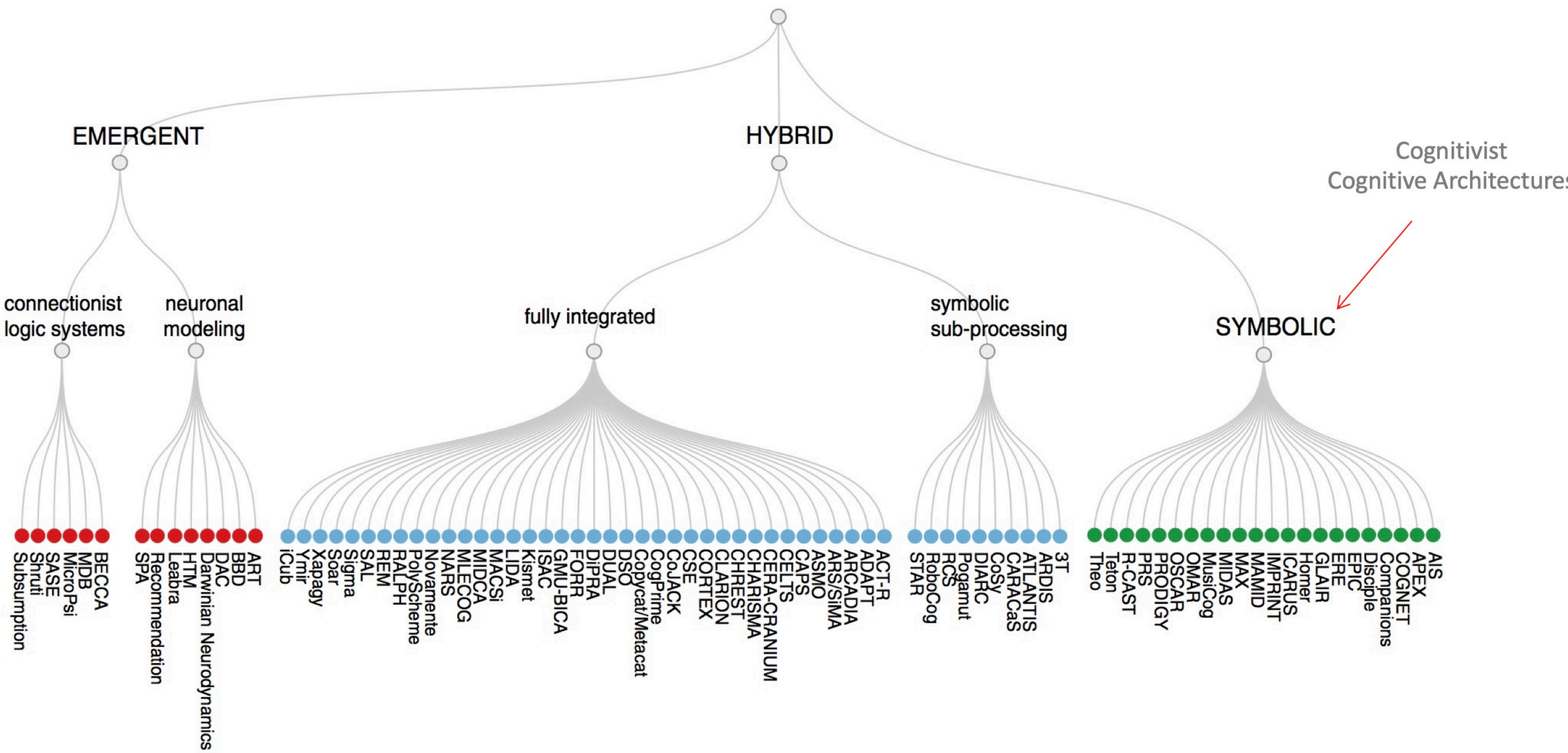
- Memories
- Formalisms for learning
- Programming mechanism

Phylogeny - basis for development

- Innate skills & core knowledge
- Memories
- Formalism for autonomy
- Formalism for development

- **Hybrid cognitive architectures** aim to combine the symbolic, rule-based processing of **cognitivist architectures** with the parallel, distributed processing of emergent architectures (i.e., *across many interconnected processing units, such as neurons or artificial nodes in a neural network*).
- For example, a **hybrid cognitive architecture** might use symbolic representations to encode abstract concepts and rules but also incorporate **neural networks** or other parallel processing techniques to support more flexible and adaptive behavior. The architecture might also include mechanisms for learning and adaptation, such as reinforcement learning or evolutionary algorithms.

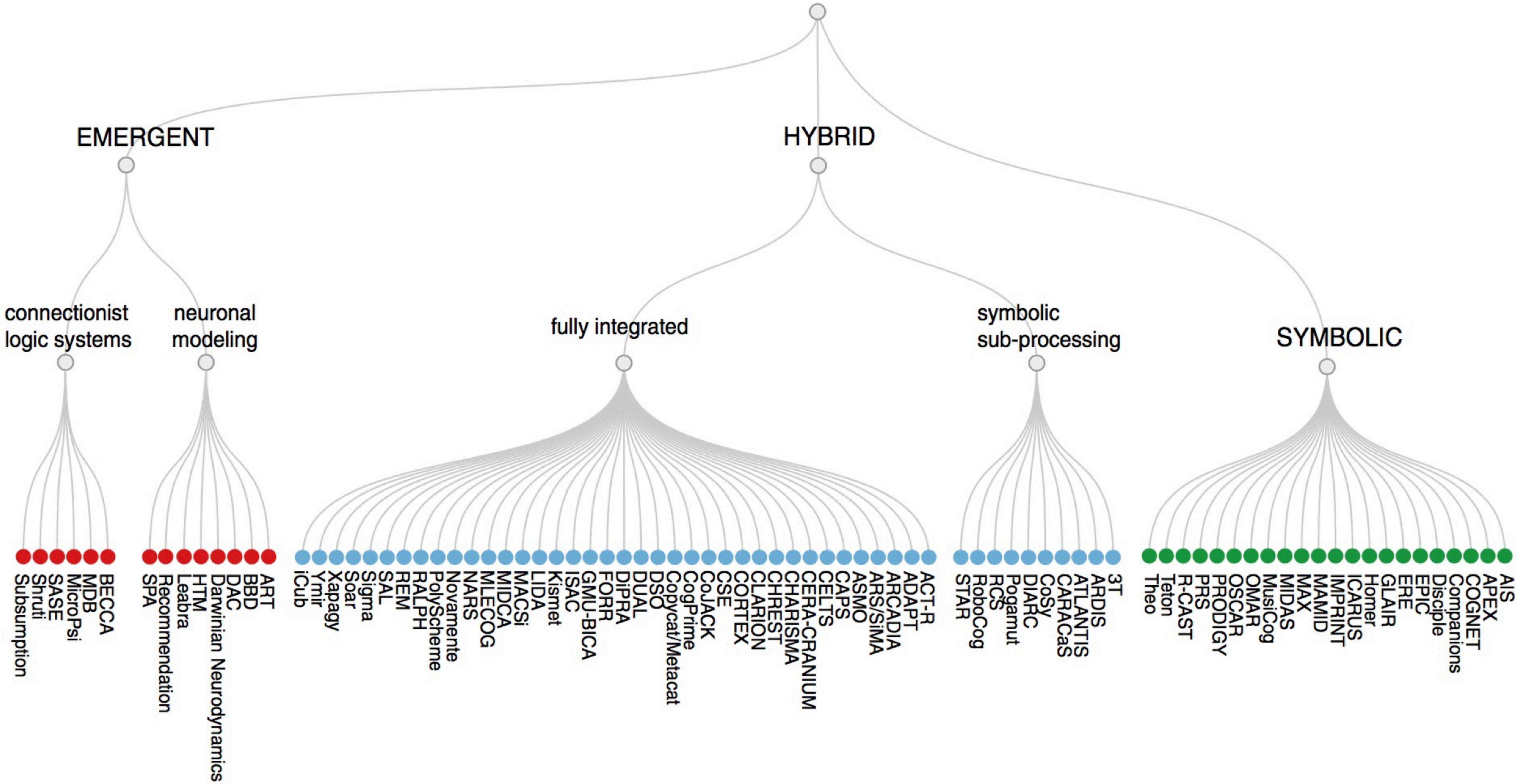
Cognitive Architectures



I. Kotseruba and J. Tsotsos. 40 years of cognitive architectures: core cognitive abilities and practical applications. Artificial Intelligence Review, Vol. 53, No. 1, pp. 17-94, 2020.

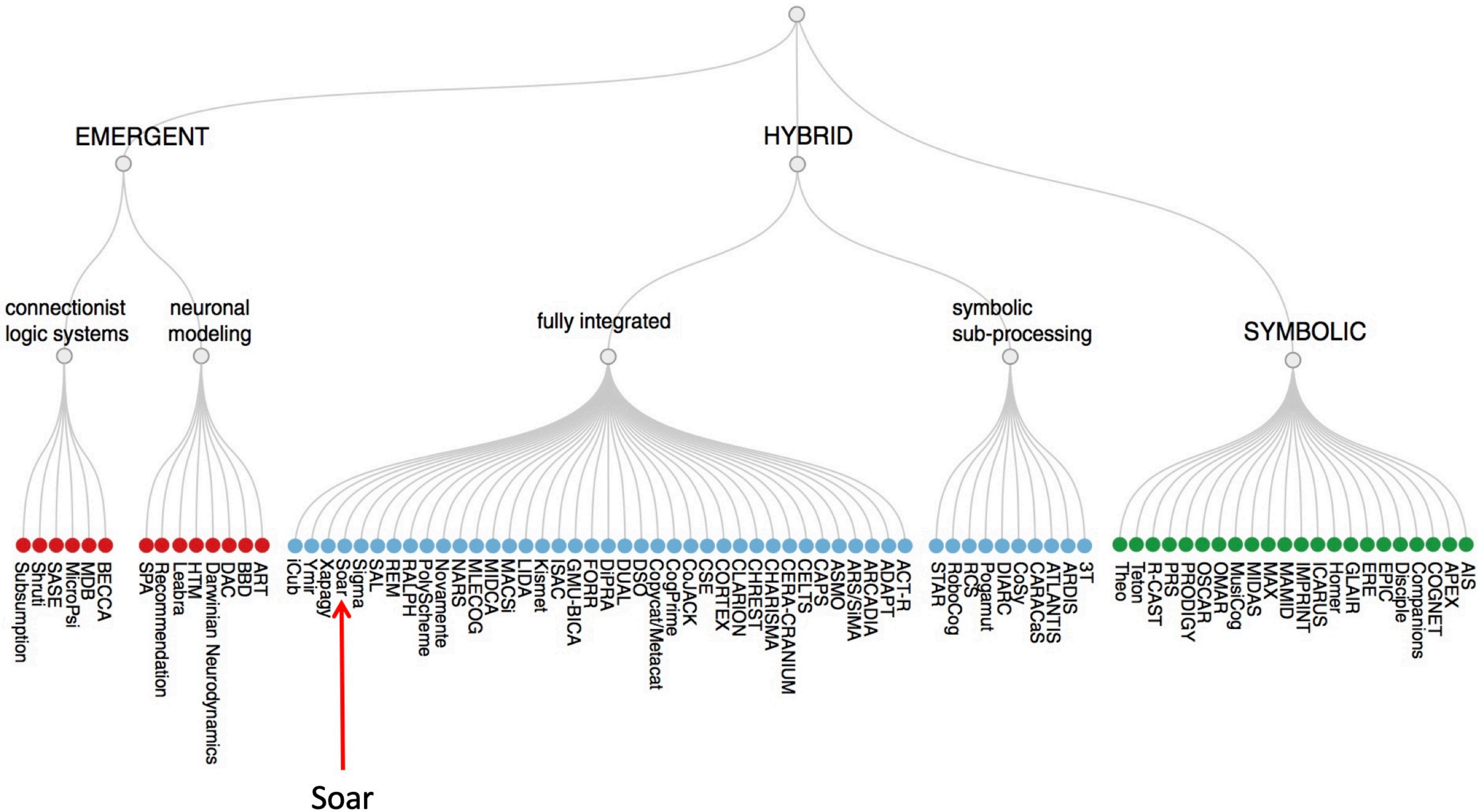
- Symbolic means knowledge is represented explicitly using rules, symbols, facts, goals, plans, and logical relations. Example: IF the robot sees a cup AND the human asks for a drink, THEN pick up the cup.
- Sub-symbolic means knowledge is not represented as clear rules or explicit symbols. Instead, it is represented through patterns, weights, activations, probabilities, or learned associations, such as in Neural Networks and Reinforcement Learning. Sub-Symbolic is suitable for applications in speech recognition, vision, control, etc.
- Fully integrated means the symbolic and sub-symbolic parts are not just placed side by side. They are connected deeply so that they support each other during cognition so perception, reasoning, learning, and action influence each other within one architecture.

Cognitive Architectures



We will briefly sample a few of the more well-known cognitive architectures

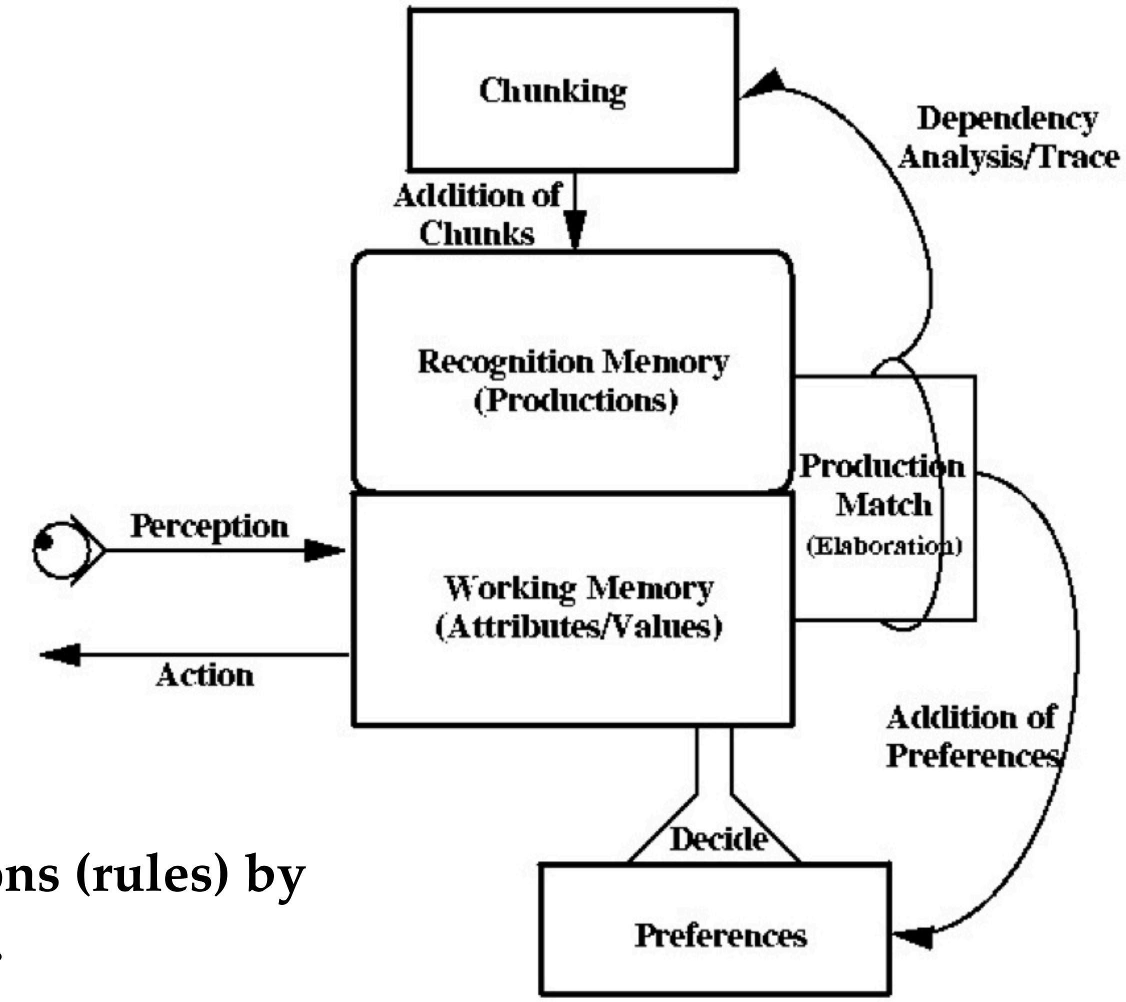
Cognitive Architectures



Cognitive Architectures

- **What does SOAR stand for?**
  - **State**
  - **Operator AND**
  - **Result**

- **An architecture for constructing cognitive models**
- **A. Newell's candidate for a Unified Theory of Cognition**
- Production (rule-based) system
- Cyclic operation
  1. Production cycle: fire all rules that match information in the symbolic working memory; update memory, fire all rules …
  2. Decision cycle: select an action
- Universal sub-goaling: create a new goal and expose more knowledge when an impasse is encountered
- Learns a new rule when an impasse is resolved

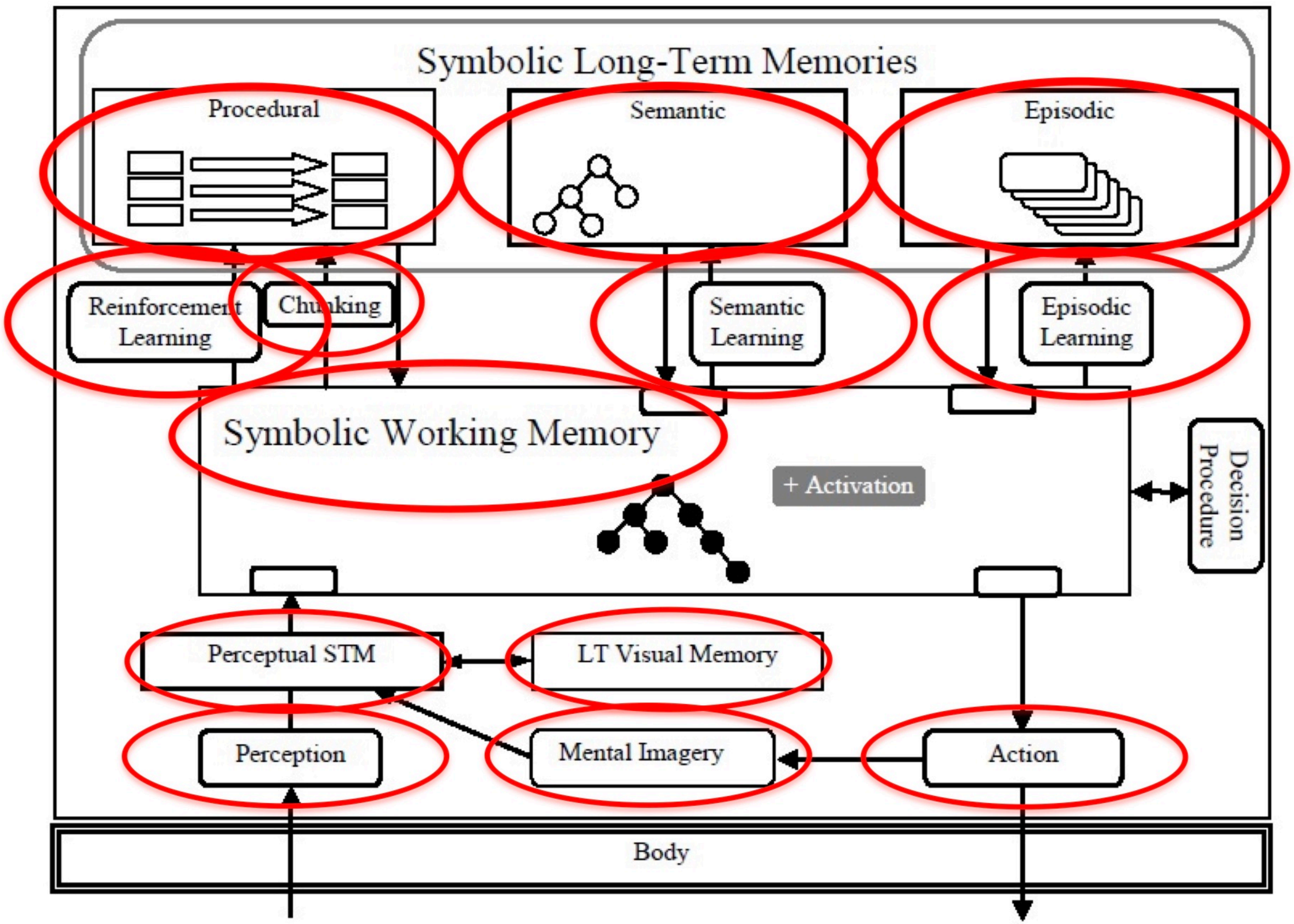


- **Chunking refers to the process of creating new productions (rules) by combining existing rules into a single, more general rule.**
- **An impasse occurs when the system cannot find a rule that applies to the current situation, or when the rules that are applicable to the current situation conflict with each other.**
- **SOAR looks at the current working memory, finds all rules that match it, and fires those rules to add useful information, propose operators, and create preferences before making a decision.**

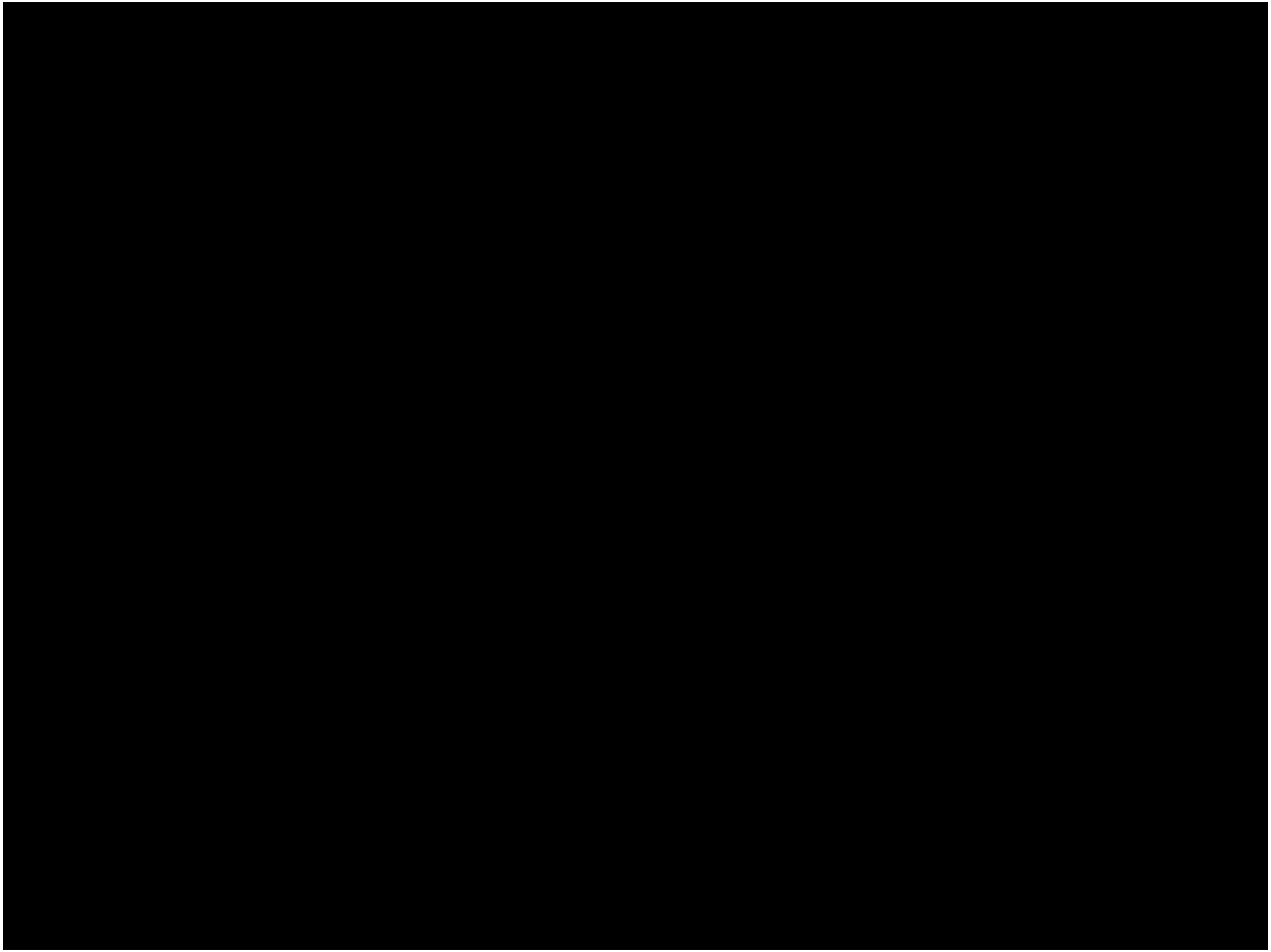
Cognitive Architectures

Soar

(Laird et al. 2012)

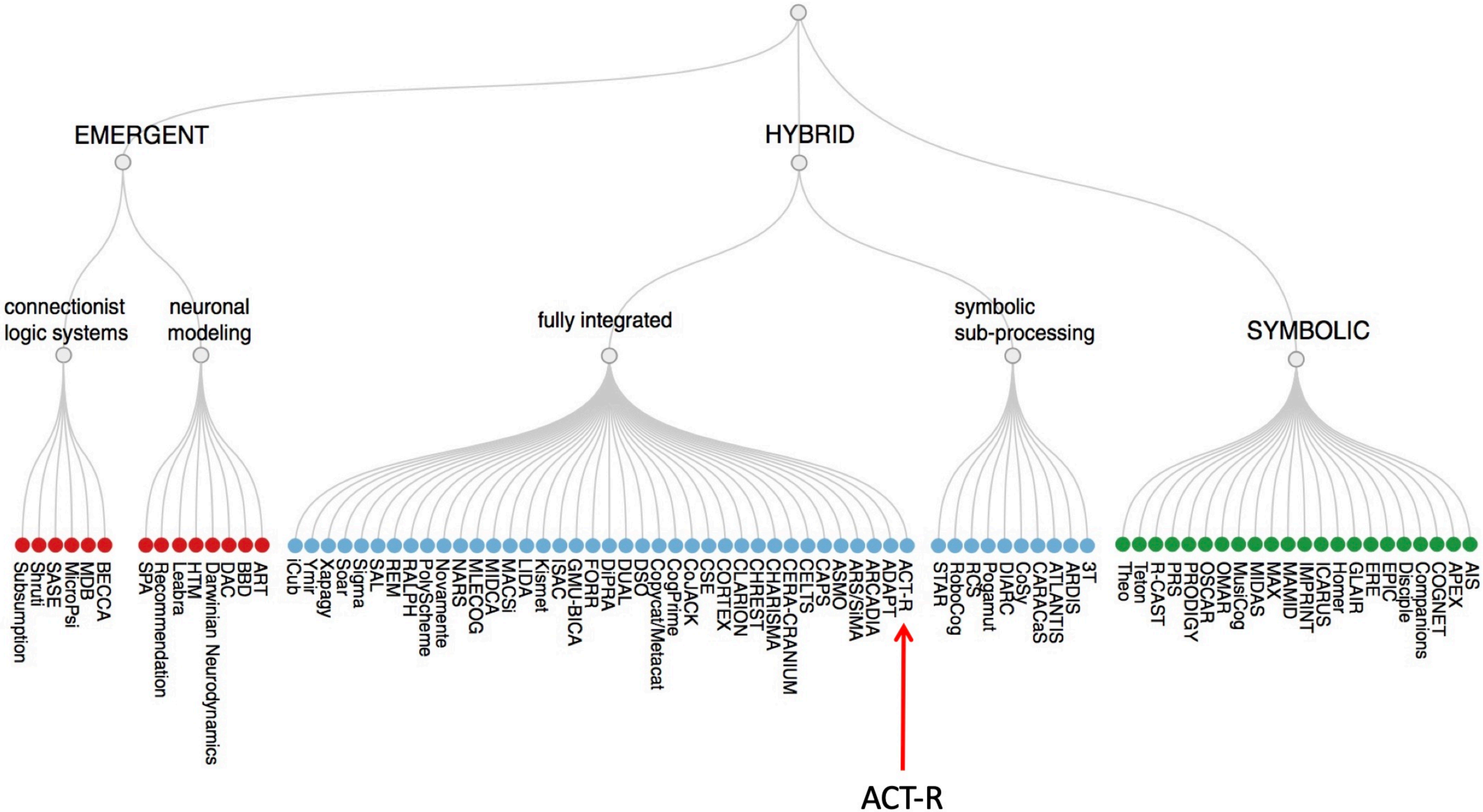


Cognitive Architectures



Full Video at: https://www.youtube.com/watch?v=2pNsfBj7XSA&feature=youtu.be

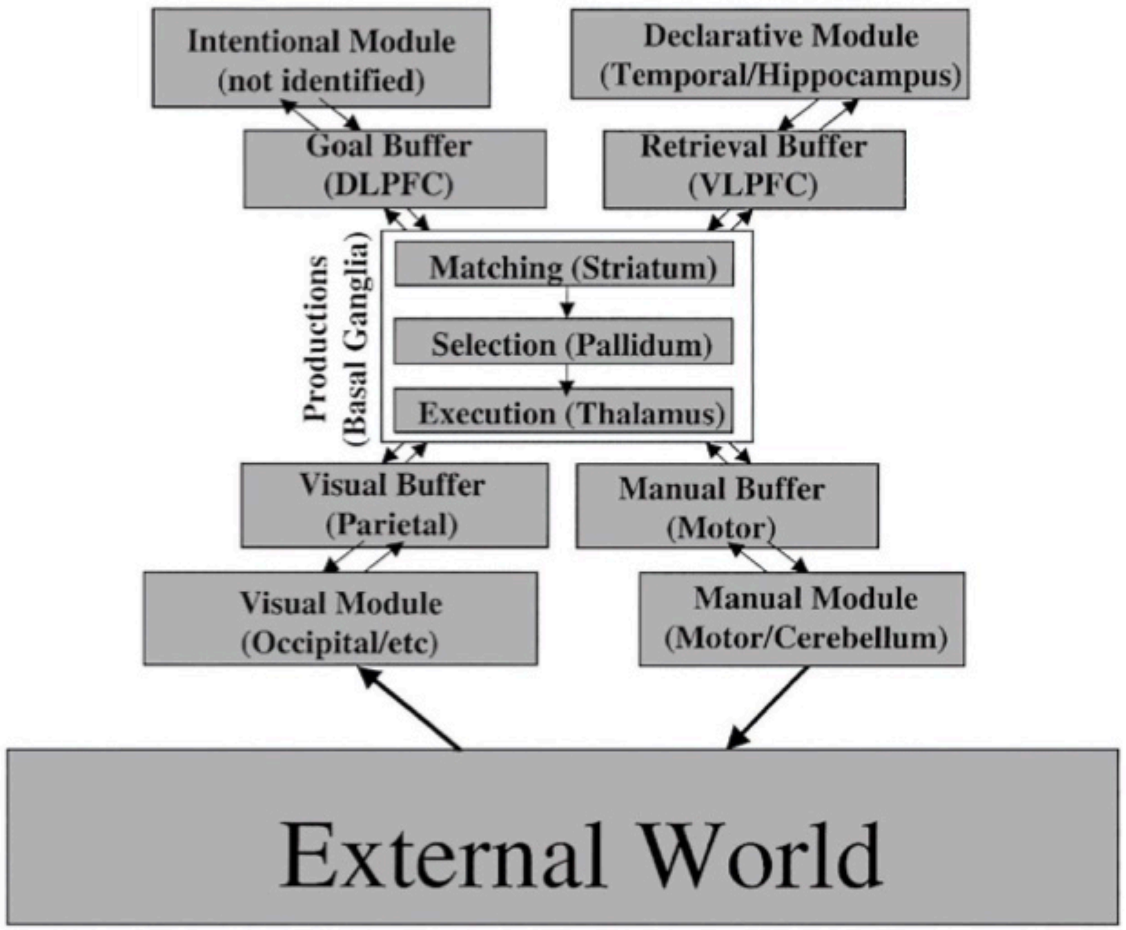
Cognitive Architectures



Cognitive Architectures

- J. Anderson's candidate for a Unified Theory of Cognition
- 1996, 2004; now version 7
- Production system with five modules: Intentional, Declarative, Visual, Manual, Production
- Cyclic operation: executes one production per cycle
    - Pattern of information in the buffers is recognized
    - A single production is selected and fires
    - The buffers are updated
- Each cycle takes approximately 50 ms.

ACT-R



- **Soar is a more general-purpose architecture that can be used to model a wide range of cognitive processes. Soar has a stronger emphasis on learning and adaptation, and includes mechanisms for acquiring new knowledge and skills through trial and error. Example of applications: Robotics, NLP, decision-making systems, etc.**
- **ACT-R is more specialised and focused on modelling specific cognitive domains such as memory, attention, and perception. ACT-R, on the other hand, is more focused on capturing existing human performance data, and is less flexible in terms of learning and adaptation. Example of applications: Cognitive psychology, neuroscience, etc. ACT-R is symbolic and rule-based.**

Cognitive Architectures

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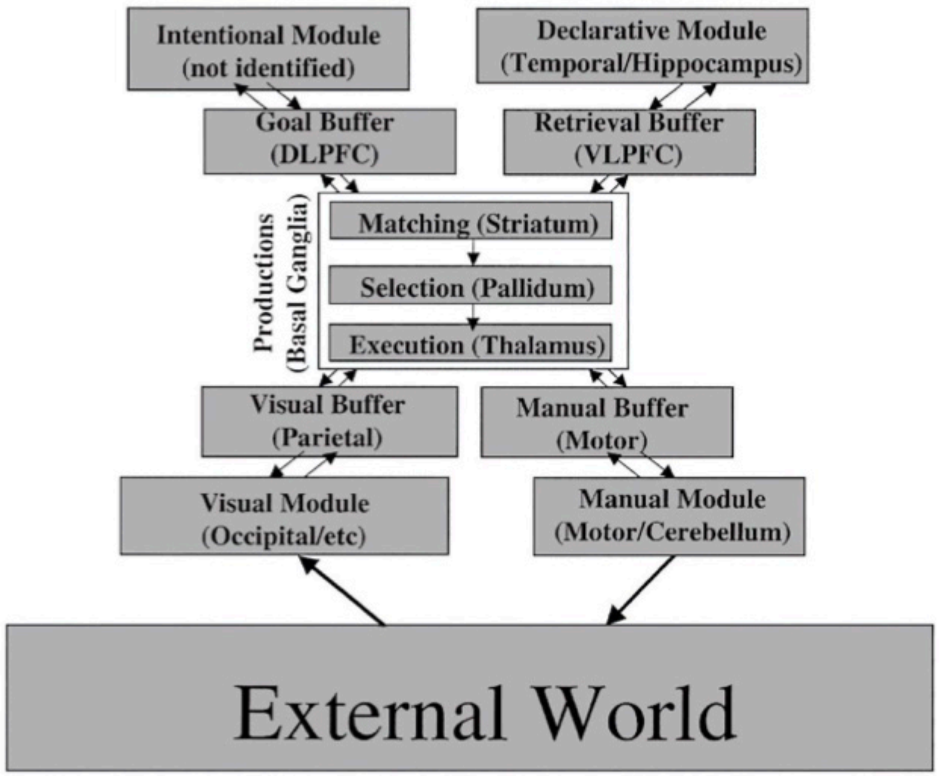
ACT-R



1. The declarative module: This module represents knowledge that is stored in long-term memory, such as facts and concepts. It is responsible for retrieving and manipulating this knowledge when needed.
2. The visual module: This module represents visual information, such as what we see in the environment. It is responsible for processing visual information and sending it to other modules for further processing.
3. The manual module: This module represents motor actions, such as reaching and grasping. It is responsible for controlling the movements of the body in response to cognitive processes.
4. The goal module: This module represents the current goals or intentions of the system. It is responsible for maintaining and updating these goals, and for coordinating the actions of other modules to achieve them.
5. The procedural module: This module represents knowledge about how to perform specific actions or tasks. It consists of a set of production rules that specify the sequence of actions to be performed in response to specific situations.

Summary

- **Today we discussed**

  1- Cognitive architectures

- **Next week**

  1- END of COMP3018/COMP5018